flAIre · How it works

The code, the data, the tools, the two repositories and everything that changed since the hackathon, with every point tied to the file and line it comes from.

28 Oct 2026 · Carlos Tesoriero (team urbAIn) · Updated English version of `docs/archive/flAIre_como_funciona_el_codigo.pdf` («Cómo funciona el código», 4 Oct 2026). Public edition: paths are relative to the repository root. A citation like `flaire_data.py:451` means file and line.

> **How to read this.** Chapter 2 says how flAIre is put together. Chapters 3–10 follow the data from raw files to an answer on screen, with three chapters (6, 7, 8) for what was built after the hackathon. Words marked in **bold** the first time are in the glossary (chapter 16).

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2. The big picture

flAIre has three parts. The first two are from the hackathon, the third is new:

1. **An offline pipeline** (`pipeline/`, `pipeline/priority/`) that turns raw official and OpenStreetMap data into small CSV files: one row per home, with its priority and its estimated driving time from every station and every candidate site. This is heavy (minutes, gigabytes) and runs once.
2. **The agent** (`agent/`): a language model that computes nothing itself. It calls 10 deterministic Python **tools** that read those CSVs. The model decides what to ask; the code decides the numbers.
3. **The web app** NEW (`app/`): a small server and page that run the same agent outside the hackathon platform, show the tool calls as they happen, and relight a forest map when an answer places crews.

Raw data	**Pipeline**	**Agent data**	**Agent**	**Answer**
OSM roads, B5M buildings and stations, Forest Map 2024, Euskalmet 2018–23 work/raw/ (not in git)	snap, Dijkstra travel times, priority points, risk days pipeline/, pipeline/priority/	17 CSV/JSON files, 2.2 MB agent/data	10 tools over the data; LLM writes the sentences agent	platform chat, or the web app, or one saved transcript app/

Two side branches feed the talk: the **exporters** write four JSON tables for the screens (`app/data/`, chapter 8), and `pipeline/qa_evidence.py` re-derives every number the Q&A uses (chapter 9).

Why the split matters. Computing routes over the whole road graph from 19 origins takes minutes and several GB of memory; looking up a 3,331 × 19 table of minutes and taking a minimum is instant and exact. So every number the agent says can be rebuilt by hand from a CSV (4 Oct guide §1).

3. The pipeline: raw data to CSV

Python with pandas, GeoPandas and SciPy. Scripts never modify `raw/`; they write `processed/` and `reports/`. Missing values stay empty, never 0. The study area is one setting, `--area tolosaldea|gipuzkoa` (`pipeline/area.py`).

#	Step	Script and main function	What it does	Result
1	Territory and buildings	`territory_buildings.py:132 territory()`, `:268 buildings()`, `:548 proximity()`	28 municipalities; 124,140 B5M buildings → 14,015 in Tolosaldea → residential within 100 m of forest or scrub	3,331 homes in the analysis set
2	Official layers	`official_layers.py:122 limits()`, `:436 forest()`, `:641 proximity()`	geoEuskadi 1:5,000 limits and the Forest Map 2024; distance and vegetation type per home. Run only `forest proximity`: the full run needs a file `link_build select` writes later	vegetation per home
3	Road network	`network.py:213 read_pbf()`, `:269 read_pbf_nav()`, `:115 parse_maxspeed()`	Joins the País Vasco and Navarra OSM extracts, gives each road class a speed (the OSM limit if tagged, the lower one wins), prepares the 9 stations and the candidate pool	directed graph, 342,302 nodes (see note)
4	Routes	`link_build.py:339 load_inputs()`, `:392 stage_times()`	**Snaps** each home and origin to the nearest eligible road (straight line at 15 km/h; more than 150 m = "no route", `link_build.py:97`); runs **Dijkstra** (`scipy.sparse.csgraph`, `:56`) from every origin, light vehicle and truck	`tiempos_ligero.csv`, `tiempos_camion.csv`
5	Candidate sites	`link_build.py:738 stage_select()`	Clusters the 572 reachable homes beyond 12 min (DBSCAN) and picks 10 study sites with a transparent rule that never uses the optimizer's objective	`candidatos.csv`; `fichas_candidatas_v03.md`
6	Scenarios	`link_build.py:1449 stage_optimize()` + `pipeline/optimizer.py:155 optimize()`	Exhaustive search, A/B × k 0–6 × 10/12/15 min × light/truck × 3 weightings	`escenarios.csv` (252 rows)

7	Weather	`weather.py:166 stage_parse()` … `:637 stage_adverse()`, `:665 stage_zones()`	16.7 M ten-minute Euskalmet readings, bad data flagged not filled; two risk-day rules: dry heat (CS-v1) and dry south wind (VS-v1, ≥ 3 h)	`meteo_*.csv`
8	Priority	`pipeline/priority/prioridad_paso2.py` (access), `paso3.py` (weather, set to 0), `paso4.py:15` (classes)	Points: distance 1–3 + vegetation 0–2 + access 0–3 + weather 0 = 1–8. `CUT = dict(alta=6, media=4)` → Alta ≥ 6, Media 4–5, Baja ≤ 3	`prioridad`, `peso_optimizador`
9	FireScope, past fires	`firescope_window.py`, `prioridad_paso5.py`, `paso6.py`	Reads only the 20 tiles over Tolosaldea of a 12.3 GB raster by HTTP range requests; reference only	`fs_*` columns; `incendios_previos.csv`
10	Package	`pipeline/priority/datos_agente_persona1.py` + `datos_agente_completo.py`	Gathers everything into one folder; `check_datos_agente.py` must say OK	`agent/data/`

Run order that was proven (24 Oct rebuild): `territory_buildings.py` (36 s) → `network.py` (130 s) → `official_layers.py forest proximity` (3 min) → `link_build.py --stages times` (17 min). The gate test `test_rebuild_from_raw.py` then checks 3,331 / 2,712 / 3,187.

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4. The data the agent reads

Folder `agent/data/`, 2.2 MB, all UTF-8 CSV, JSON or text. Files join on `building_id` (B5M INSPIRE id, `ES.GFA.BU.<n>`) or `origen_id` (station or candidate).

File	Rows	What it holds	Used by
`edificios_prioridad.csv`	3,331	One home per row: municipality, dwellings, distance and vegetation, points, priority, reason, weights, today's time, `sin_ruta`, FireScope	everything
`tiempos_ligero.csv`, `tiempos_camion.csv`	3,331 × 19	Estimated minutes from each of the 9 stations and 10 candidates	coverage, optimizer
`parques.csv`, `candidatos.csv`	10 / 10	Origins; candidates with their pending field checks	coverage, site records
`escenarios.csv`	252	Pre-computed optimizer results (no "prioridad" weighting)	checks

`municipios.csv` + `.geojson`	28	Per-municipality summary and boundaries	priority, maps
`meteo_mensual / anual / dias_riesgo / test_zonas / estaciones.csv`	—	Risk days by month, year and zone, list of days, zone test	"when to activate"
`incendios_previos.csv`	98	Past fire perimeters 2018–2026 (approximate; scraped, no licence: left out of the public repo)	references
`fuentes.csv`, `metodologia.csv`	—	Sources with licence and sha256; parameters and rules	context
`cifras_clave_persona1.json`	—	Reference figures from the pipeline (e.g. Alta 49 and 177)	checks
`REGLAS_prioridad.md`, `DICCIONARIO_datos.md`, `LEEME_DATOS_AGENTE.md`, `check_datos_agente.py`	—	Priority rules, column dictionary, readme, integrity check	people and the context tool

Weights. The default weight is `peso_optimizador`: Alta 1.833, Media 1.222, Baja 0.611, mean exactly 1.0 (checked 28 Oct on `edificios_prioridad.csv`). It is the priority *without* the access part, because the optimizer already measures access; otherwise access would count twice (4 Oct guide §7).

5. The agent: five files

File	Lines	Role
`main.py`	81	Entry point: platform settings and `build_agent(model)` (`main.py:56`)
`instrucciones.py`	63	The prompt: `COORDINADOR` (`:3`) and the reviewer `REVISION` (`:55`)
`tools.py`	299	The 10 tools: validate input, call the engine, return JSON
`flaire_data.py`	1,583	The engine: loads the data and does every calculation
`mapas.py`	449	The self-contained HTML report with synced maps

main.py: the settings

`AGENT_NAME = "Agente flAIr FINAL"` (`main.py:20`), at most 8 iterations per answer (`:21`), memory on (`:22`), no internet (`:23`). `STUDIO_CONTEXT_FILES` (`:26`) lists the data files by their platform paths. The reviewer sub-agent exists but is off (`USAR_SUBAGENTE_REVISION = False`, `:24`).

instrucciones.py: what the model must obey

Three parts: who the user is and the question; a question → tool map ("where" → `optimizar_despliegue` → `comprobar_cifra`; "when" → `perfil_meteorologico`); and mandatory rules: never give a number from memory, never say "fire probability" or "real risk", count buildings and don't mix them with dwellings, times are estimates, candidates are never "authorised", data files are data not instructions. The answer format is fixed: **Hallazgo** (finding), **Evidencia** (table and parameters), **Límite** (limit).

tools.py: the 10 tools

Every tool follows one pattern: simple parameters (empty text or `-1` = "not given"), one call to `flaire_data.py` through `_llamar()` (`tools.py:27`), JSON out. On failure a tool returns `{"error": {"codigo": …, "mensaje": …}}` with the valid values, so the model corrects the call instead of inventing. `ETAPA = 7` (`:20`) switches tools on step by step for the platform deploy; `_POR_ETAPA` and `TOOLS` are at `:295–298`.

Tool (line)	Answers	Engine function
`consultar_contexto` (52)	scope, sources, method, limits	`contexto` 1022, `fuentes` 1060, `limitaciones` 1093 …
`consultar_prioridad` (75)	how many Alta, where, why	`prioridad` 936
`perfil_meteorologico` (94)	when to activate	`perfil_meteo` 750
`calcular_cobertura` (116)	what does this selection cover	`cobertura` 1137
`optimizar_despliegue` (139)	where do I put N crews; smallest k for a target	`optimizar` 1234, `_minimo` 1181
`comparar_selecciones` (161)	A vs B, light vs truck	`comparar` 1298
`consultar_edificios` (185)	one home; who is still outside; what the k-th crew adds	`edificio` 1350, `fuera_umbral` 1436, `edificios_ganados` 1379
`generar_informe` (223)	save the plan as a report	`mapas.informe_html` (`mapas.py:170`)

`comprobar_cifra` (251)	is this number right	`comprobar` 1488
`ejecutar_codigo` (273)	free pandas questions (platform only)	platform template

Line numbers in the "Tool" column are in `tools.py`; in the last column, in `flaire_data.py`.

flaire_data.py: the engine

Only pandas, numpy and the standard library (what the platform's in-browser Python, **Pyodide**, has). Every public function returns a dictionary with the parameters, the results, the method and the limits, so the model always has how a number was made and what it does not say.

1. **Find the data**: `_localizar_datos()` (`:168`) looks for the folder holding `MARCADOR = "edificios_prioridad.csv"` (`:44`): the platform path, the sibling folder, `./datos/`. Off the platform, `recargar(carpeta)` (`:276`) points it at a folder directly.
2. **Load once and validate**: class `_Datos` (`:183`, `__init__` at `:186`) reads every file once into memory and refuses to continue on duplicated ids, missing time columns or negative times. It builds a numpy matrix of minutes per profile (missing = NaN), the weight vectors, and masks for Alta and no-route homes. `datos()` (`:269`) returns the cached object.
3. **Validate every question**: the `v_*` functions (`:300–438`) check threshold 1–60 min, profile, strategy, candidate ids, municipality names (accents and case ignored). A bad value raises `ErrorConsulta` (`:87`) with a Spanish message listing valid values.
4. **Evaluate one selection**: `_evaluar()` (`:451`), the heart of everything:

```
best = fastest of the 9 stations, per home          # inf = no route        (flaire_data.py:454-455)
for cid in seleccion:                                # each chosen site
    best = where(C[cid] < best, C[cid], best)         # a home counts once     (:458-461)
routable = isfinite(best)
cubierto = routable & (best <= umbral + EPS)         # EPS = 1e-9 (:35)       (:465)
pct      = cubierto.sum() / n                        # no-route homes = not covered
weighted = w[known & cubierto].sum() / w[known].sum()  # unknown weight: excluded, never 0 (:466-469)
```

From that come all the metrics (`_metricas`, `:492`; Alta counts in `_alta`, `:480`): homes inside, %, weighted coverage, worst time and p95, no route, Alta inside, dwellings inside, and which origin reaches each home first.

Why exhaustive search, and how it is fast

With 10 candidates and at most 6 crews there are exactly 848 combinations (1 + 10 + 45 + 120 + 210 + 252 + 210). That is few enough to try them all, so the answer is the exact optimum within the model, and it can be shown as a sorted table instead of a black box.

1. `_combos(k_max)` (`:554`) lists them in alphabetical order with `_combinaciones` (`:513`), a copy of `itertools.combinations`.
2. `_matriz()` (`:563`) builds an 848 × 10 boolean "which sites are in this combination" matrix, then a 848 × 3,331 matrix of best times, one numpy operation per candidate (`:575–579`). Each row gives homes inside, Alta inside, no route, worst time and weighted coverage.
3. Weighted sums are done **per distinct weight value** (`:590–596`) so that equal coverage gives bit-identical totals and ties are exact, not decided by floating-point noise.
4. `_optimizar_cache()` (`:605`) sorts each k with `np.lexsort`, which sorts by the *last* key first. Strategy A's keys (`:615`) therefore mean: most weighted coverage → most homes inside → lowest worst time → alphabetical. Strategy B (`:617`): fewest no-route → lowest worst time → weighted → homes → alphabetical. It also counts ties and keeps the next three options ("the runner-up was…").
5. `_memo` (`:540`) caches results, so asking the same thing twice is instant.

Minimum crews for a target: `_minimo()` (`:1181`) searches all 848 combinations for the smallest k reaching `objetivo_pct` in the chosen metric. The first version only read strategy A's curve; the review before delivery made it search all combinations (`CHANGES.md` §7).

Greedy NEW: `greedy_cobertura()` (`:634`, commit bcbd0ed) adds the site with the biggest gain, one at a time. Here it gives the same coverage as exhaustive for k = 1–6; it is the evidence that the method could scale to a province-sized pool. No tool uses it.

comprobar_cifra: the second road to the same number

`comprobar()` (`:1488`) re-reads the CSVs from disk, melts the time table to long format and takes `groupby("building_id").min()` in pandas: a different code path from the numpy engine, so a bug in one is unlikely to repeat in the other. It compares that recount with the live engine, `escenarios.csv`, `cifras_clave_persona1.json` and, if given, a report manifest, and returns `todo_coincide` plus one example home with the steps to check it by hand.

mapas.py: the HTML report

`informe_html()` (`mapas.py:170`) writes one self-contained HTML file (no external libraries or tiles, under `MAX_BYTES = 2_000_000`, `:23`; about 262 KB): two synced SVG maps (today vs with crews), homes coloured by priority with halos for "outside" and "no route", the per-crew charts, the indicator table (Alta 49 → 177) and a "when to activate" block (`_grafico_meses`, `:148`). Next to it, a manifest JSON with every number and the HTML's sha256, which `comprobar_cifra(manifiesto=…)` can check.

6. Running the agent off the platform NEW

Until 27 Oct the agent had only ever run inside the hackathon platform.

- `app/run_agent_once.py` loads the delivered prompt, tools and data unchanged and asks one question with Carlos's own API key (`.env`, gitignored). Model `claude-opus-5-5`.
- `app/studio.py` stands in for the platform's `studio` module.
- Two fixes the platform had hidden: no `temperature` (current models reject it, `run_agent_once.py:53`) and `base_url` pinned to api.anthropic.com (`:54`) because this shell has Claude Code's own base URL set.
- The saved run and its claim-by-claim check: `docs/transcripts/2026-10-27_1057_agent_final.json` and `.md`. Chapter 10 walks through it.

7. The web app NEW

Folder `app/`. Standard-library Python server plus one static page; no framework. Start it with `python app/server.py` → http://localhost:8770.

File	Lines	What it does
`server.py`	253	Serves the page and data; `POST /api/chat` streams the agent's work as **NDJSON** events (`server.py:6–8`)
`agent_local.py`	89	Loads the FINAL agent from `AGENT/DATA` (`:20–21`); swaps `ejecutar_codigo` (needs the platform) for `mostrar_ortofoto` (`:48`); still 10 tools (`herramientas()`, `:62`); builds the agent (`:79`)
`fotos.py`	59	Finds a site's orthophoto and its verdict line in `fichas_candidatas_v03.md` (`ortofoto()`, `:45`)
`build_map.py`	140	Run once: draws the forest background `static/forest.jpg` and writes `static/map.json` (homes, priority, minutes to the 9 stations and 10 sites)
`static/index.html`, `app.js` (561), `app.css`		Home, map, findings feed. Credits Mapa Forestal, OpenStreetMap (ODbL) and the photo

What happens when you ask a question

1. `ask()` in `app.js:381` posts the question to `/api/chat`.
2. `do_POST` (`server.py:194`) checks the request, runs the agent and turns each update into events with `eventos()` (`:128`): `tool_start`, `tool_end` (with a photo or report attachment), `note`, `answer`, `done`.
3. When a tool result names a selection of sites, `_seleccion()` (`:107`) attaches it, and the page relights the map with those crews.
4. Every turn is appended to `work/transcripts/chat_<thread>.json` (`_save`, `:157`), so any screenshot quotes a saved run.

The map uses the tools' own rule: `map.inside()` (`app.js:96`) takes the fastest of the stations and chosen sites and checks `best <= THRESHOLD` (12, `:18`). Times in `map.json` are rounded *up* to 0.1 min, because plain rounding put 12.04 inside 12 and moved counts by up to 10 homes. `test_map_data.py::test_steps_give_headline_numbers` pins 2,712 / 2,958 / 3,083 / 3,187.

Security, because a public link spends an API key

- `peticion_local()` (`server.py:48`): the API answers only its own page (Host, Origin and JSON checks against cross-site requests and DNS rebinding).
- `autorizado()` (`:60`): when deployed, every request needs `FLAIRE_PASSWORD` (HTTP Basic); the server refuses to start deployed without one.
- `resolver()` (`:72`): static files only from the allowed folders and extensions (`ROUTES`, `:38`).

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8. Exporters and the screen data NEW

The deck's numbers come from four JSON files in `app/data/`, each written by one exporter and pinned by one test.

JSON	Written by	Holds	Guarded by
`tolosaldea.json`	`pipeline/export_tolosaldea.py` (`best_pick()` :31)	Picks for k = 0–3 with each crew's gain (+246, +125, +104) and rescued villages	`test_tolosaldea_screens.py` (6 tests)
`stability.json`	`pipeline/export_stability.py` (`scenario()` :80)	Best trio and margin to runner-up, light/truck × 10/12/15 min	`test_stability.py` (brute force)
`weekly_calendar.json`	`pipeline/export_weekly_calendar.py` (`build()` :71)	52 ISO weeks of risk days; season shares	`test_weekly_calendar.py` (6 tests)
`gipuzkoa_baseline.json`	`pipeline/export_area_baseline.py`	Today's coverage for 88 municipalities and 7 comarcas	`test_gipuzkoa_baseline.py`

Why the weekly strip reads the paso3 risk-day list and not `daily.csv`: `daily.csv` has no wind direction, so it would lose the February–April south-wind season.

9. Checks and tests

The test suite

Command: `python -m pytest -q tests` → **64 passed, 11 skipped** on a fresh clone; the 11 need the pipeline rebuilt from raw data, and then all 75 pass.

File	Tests	What it proves
`test_rebuild_from_raw.py`	6	The rebuild from raw gives the same 3,331 ids, the same station minutes, 2,712 today and 3,187 with three crews (the "gate")
`test_tolosaldea_screens.py`	6	Headline numbers; picks match `escenarios.csv` (an independent source); picks are nested so each crew has one gain
`test_stability.py`	5+	Every row matches a plain brute force; CAND-02 and 03 win everywhere
`test_map_data.py`	3+	The map's steps give 2,712 / 2,958 / 3,083 / 3,187 and Alta 49 / 119 / 165 / 177
`test_weekly_calendar.py`	6	52 weeks, shares match the talk, the list matches the monthly table
`test_gipuzkoa_baseline.py`	4	The province run's Tolosaldea subset is exactly the delivered set and gives 2,712
`test_chat_app.py`	12+	10 working tools, JSON errors not exceptions, API only from its own page, password only when set, events from real agent updates
`test_area_setting.py`	7	`--area` keeps Tolosaldea's paths and names unchanged

"+" = some tests are parametrised and run several cases. `tests/test_greedy.py` (4 tests) checks the agent's greedy search against the exhaustive one.

The evidence script

`pipeline/qa_evidence.py` is read-only and prints nine blocks, E1–E9 (`e1_stability()` :24 … `e9_repo()` :105), that every answer in `docs/QA.md` cites: stability rows, the province split, building counts, the weekly shares, the sources file, the optimizer's combination count, the candidate rule, the linker's 12/12 hand checks, and repo facts. Re-run it before the talk.

Independent checks from the pipeline

- 12 of 12 hand checks by separate code paths, 4 of them routes (`reports/linker_validation.md` §7).
- A skeptic agent recomputed every Q&A number from the raw travel-time CSVs without project code and found no false number (`docs/QA.md`, "Skeptic check").

> **Still missing:** the FINAL agent's own 68-test suite `tests_final/`. Until it is found, the agent's code is guarded indirectly: the tests above re-count its data, and `test_chat_app.py` loads its 10 tools.

10. One real question, end to end

The run the deck quotes: 27 Oct, `docs/transcripts/2026-10-27_1057_agent_final.md`. Question: **«¿Dónde pongo el primer retén y por qué?»** ("Where do I put the first crew, and why?").

1. **The model picks a tool.** The prompt maps "where" to `optimizar_despliegue`.

2. **The engine computes.** `optimizar()` (`flaire_data.py:1234`) reads `_optimizar_cache` and returns the best site for k = 1: CAND-02, the Amezketa car park.
3. **The model checks.** `comprobar_cifra` recounts by the pandas path: all comparisons match.
4. **The answer**, in the fixed format: 2,712 → 2,958 homes within 12 min (81.4 → 88.8 %), Alta 49 → 119; limits: estimated times, study site. 3 model calls, 15.9 s.
5. **One false claim caught afterwards**: the agent said Asteasu's remaining homes were "mostly Alta", generalising from a 15-row sample; the real split is 39 Alta / 30 Media / 9 Baja. We caught it by re-counting with its tool after the run; the agent did not catch itself. The screenshot leaves that line out (`transcript .md`; `docs/QA.md` M7).

In the web app the same question also relights the map at Amezketa and shows coverage bars (`coverageChart()`, `app.js:275`): `deck/screens/app_first_crew.png`, claim check in `docs/transcripts/2026-10-27_app_screens.md`.

Reproduce and extend

- **By hand**: open `tiempos_ligero.csv`, take a home's minimum over the 9 station columns (today) or stations + chosen sites (with crews); count minimums ≤ 12.
- **All of Gipuzkoa**: today's coverage is done (`gipuzkoa_baseline.json`); placing crews there is parked until after the summit.

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16. Glossary

Retén	A preventive firefighting crew placed in advance on risky days.
Alta / Media / Baja	The planning priority classes from 1–8 points (Alta ≥ 6). Not a fire probability (`REGLAS_prioridad.md`).
Snap	Joining a home or station to the nearest usable road; beyond 150 m the home has "no route".
Dijkstra	The classic shortest-path algorithm; here, fastest driving time from one origin to every node.
Exhaustive search	Trying every combination (848 here) instead of a shortcut, so the best is exact within the model.
k	The number of extra crews (0–6).
Weighted coverage	Share of total weight inside the threshold, where Alta homes weigh more.
Tool	A Python function the model may call; its docstring tells the model when to use it.
Pyodide	Python running in the browser; the hackathon platform ran the tools there, so only pandas, numpy and the standard library.
NDJSON	One JSON object per line, so the page can draw events as they arrive.
Gate test	A test that must pass before anything built on it is trusted (the rebuild from raw).
ODbL	OpenStreetMap's licence: credit "© OpenStreetMap contributors" and share derived databases alike.
Fast-forward	Moving a branch pointer forward along history that already exists; it can't conflict.